Hema Goud

Data Scientist Aspirant

Hyderabad

Summary

Motivated AI/ML Fresher with hands-on project knowledge in Machine Learning, Deep Learning, NLP, and Generative AI. Skilled in Python and passionate about developing intelligent, data-driven applications while continuously learning new AI technologies.

Education

Data Science
Innomatics Research Science
09/2025 | Hyderabad, India
Coursework in Python, Statistics, Data Analysis, SQL + PowerBI, ML, DL, GenAI, NLP, Computer Vision

BSC Forensic Science
Kalinga University
06/2022 – 05/2025 | Raipur, India

Skills

- **GEN AI** — LLMs, Prompt Engineering, LangChain, OpenAI API, Hugging Face,RAG, Fine Tuning
- **Deep Learning** — Neural Networks, ANN, RNN, LSTM, BiLSTM, TensorFlow, Keras, NLP, Model Evaluation
- **Machine Learning** — Regression, Classification, Clustering, Cross Validation, Hyperparameter Tuning
- **PYTHON** — Python, NumPy, Pandas, Matplotlib, Seaborn, Scikit learn

Languages

English | Hindi | Telugu

Certificates

- GEN AI | DEEP LEARNING | MACHINE LEARNING | PYTHON | EDA | POWER BI | SQL |
  Certification at Innomatics Research Labs

Projects

AI Conversational Chatbot using LangChain & OpenAI GPT

- Developed an AI conversational chatbot using LangChain, OpenAI GPT, and ConversationBufferMemory to generate context-aware, multi-turn responses.
- Implemented Prompt Engineering, LLMChain, and custom prompt templates to deliver accurate and dynamic conversational experiences.
- Integrated the ChatOpenAI API with memory management and output parsers for seamless real-time interactions.
- Built and deployed an interactive Streamlit application, enabling continuous user conversations with persistent chat history.

Fake News Detection using Recurrent Neural Network

• Developed an end-to-end Fake News Detection System using RNN and NLP, achieving 85% classification accuracy on news article data.
• Preprocessed and analyzed 40K+ news articles through text cleaning, tokenization, sequence generation, and feature engineering.
• Implemented Tokenizer, Padding, Embedding, and RNN layers using TensorFlow/ Keras to capture contextual patterns in textual data. Evaluated model performance, improving fake news identification reliability.
• Deployed the model using Streamlit, enabling real-time predictions through an interactive web application.

Mental Health Text Classification

• Developed an end-to-end Mental Health Text Classification System using Bidirectional LSTM and NLP, achieving 80% classification accuracy across 7 mental health categories.
• Preprocessed and analyzed 50K+ text records through text cleaning, label encoding, tokenization, sequence generation, padding, and class imbalance handling using class weights.
• Implemented Tokenizer, Embedding, Bidirectional LSTM, and Softmax layers using TensorFlow/Keras to capture contextual patterns in textual data, improving the accuracy of multi-class mental health prediction.
• Deployed the model using Streamlit, enabling real-time mental health classification through an interactive web application.